

Dr. Paul Schmidt

DATA SCIENTIST / BIOSTATISTICIAN

Hamburg, Germany

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📄 ResearchGate



Professional Experience

Data scientist / General Manager

Hamburg, Germany

BioMath - Applied Statistics and Informatics in Life Sciences

Since 2019-01

- End-to-end analytics: clarify questions, acquire/align data, build models, translate results for both technical and executive audiences
- Selected projects: time-series & regression on air parameters; agricultural treatment comparisons; monitoring design & evaluation; epidemiological meta-analyses; spatial analysis with GIS
- Scaled SOPs (e.g., for systematic reviews & meta-analyses): consolidated toolchain, added automation, improved reproducibility and cut cycle time
- Systematic reviews led; authored, edited, and finalized scientific reports/publications
- General manager since 2022

Workshop teacher

see 'Workshops' section below

Freelancer (part-time)

Since 2018-11

- Develop and teach workshops on statistics in R; specific content and language according to the contractor
- Provide corresponding material on my websites (see 'Other skills' section below)

Research associate

Stuttgart, Germany

University of Hohenheim

2015-09 - 2018-12

- Personalized consulting (ranging from single-appointment to project-accompanying) for students and research associates in terms of experimental design, data handling, statistical analyses and/or presentation of results
- Develop, conduct and manage yearly statistical analysis of yield stability data for external company
- Develop, organize and teach workshops in statistics, R and SAS
- Supervise student writing his MSc thesis

Junior Data scientist

Rostock, Germany

BioMath - Applied Statistics and Informatics in Life Sciences

2015-01 - 2015-08

- Streamline statistical analyses of monitoring data
- Implement SOP for systematic literature reviews

Education

Dr. sc. agr.

Stuttgart, Germany

University of Hohenheim

2015-09 - 2019-11

- DFG-funded PhD student in the biostatistics unit of Prof. Dr. Hans-Peter Piepho
- Cumulative doctoral thesis: 'Estimating heritability in plant breeding programs' graded 'magna cum laude'

Visiting PhD student

West Lafayette, IN, USA

Purdue University

2015-09 - 2015-12

- Visiting PhD student in the statistical bioinformatics unit of Prof. Dr. Rebecca Whitbeck Doerge
- Arranged on personal initiative, this collaboration allowed for scientific exchange and inspiration at the beginning of my PhD

MSc Crop Science: Plant Breeding

Stuttgart, Germany

University of Hohenheim

2012-10 - 2014-12

- Specialization in biostatistics and plant breeding (final grade 1.4)
- MSc Thesis: 'Statistical Evaluation and Analysis of PACTS trials as a series of on-farm strip trials without replicates' graded 1.0

BSc Agribiology (in german)

University of Hohenheim

Stuttgart, Germany

2009-10 - 2012-09

- Specialization in plant sciences and genetics (final grade 1.9)
- BSc Thesis: 'Cumulative effects of glyphosate trace concentrations during root exposition of winter wheat' graded 1.0

Student exchange year

Alexander Central High School

Taylorsville, NC, USA

2006-08 - 2007-07

- Completed senior year and obtained high school diploma

Skills

Analytics & Statistics

(Generalized) Linear/Mixed Models, Exploratory & descriptive analysis, Experimental design, Machine Learning

Communication & Impact

Executive-ready insights, Data visualization, Reports/scientific publications, Talks/workshops

General

collaboration, communication, structuring, time management, strategic oversight, problem solving

Languages

German (native), English (fluent)

Open Source

Website https://schmidtpaul.github.io/dsfair_quarto/, R package BioMathR <https://schmidtpaul.github.io/BioMathR/>, R package CitaviR <https://schmidtpaul.github.io/CitaviR/>

Software & Tools

R, Python, SAS, SPSS, LLM-assisted workflows, Quarto/Markdown, Markdown, Version Control (git), reproducible pipelines & DAG workflows, SQL, MS Office (VBA)

Workshops

2025

- 2025 Dec — Statistics with R - an Introduction — Universität Bonn via zoom
- 2025 Nov — exp. Design - Practicals in R — CIHEAM Zaragoza via zoom
- 2025 Nov — Data Science with R - an Introduction — Max Planck Inst. Tübingen via zoom
- 2025 Oct — Data Analytics mit Python — Karrierecenter der Bundeswehr (e-learning)
- 2025 Jul — Statistics with R - an Introduction — Universität Bonn via zoom
- 2025 May — Statistics with R (Beginner) — Universität Kassel via zoom
- 2025 Apr — R-Statistics for Beginners — Pro-RUWA (e-learning)
- 2025 Apr — Data Analytics mit Python — Karrierecenter der Bundeswehr (e-learning)
- 2025 Mar — Data Scientist - Focus Python — Karrierecenter der Bundeswehr (e-learning)
- 2025 Mar — Data Analytics mit Python — Karrierecenter der Bundeswehr (e-learning)
- 2025 Mar — Data science for exp. life sciences with R (pt. 2) — Forsch.Einr. BMEL via zoom
- 2025 Mar — Data Science in den exp. Naturwiss. (TI. 2) — Forsch.Einr. BMEL via zoom
- 2025 Mar — Data science for exp. life sciences with R (pt. 1) — Forsch.Einr. BMEL via zoom

- 2025 Feb — Data Science in den exp. Naturwiss. mit R (Tl. 1) — Forsch.Einr. BMEL via zoom

2024

- 2024 Dec — Statistics with R - an Introduction — Universität Bonn via zoom
- 2024 Oct — Data Analytics mit Python — Karrierecenter der Bundeswehr (e-learning)
- 2024 Sep — Data science with R for scientists — Universität Rostock via zoom
- 2024 Sep — Statistics with R — Universität Hamburg via zoom
- 2024 Sep — R and the {Tidyverse} — TU Dortmund
- 2024 Aug — Data Analytics mit Python — Karrierecenter der Bundeswehr (e-learning)
- 2024 Jul — Statistics with R - an Introduction — Universität Bonn via zoom
- 2024 Jun — Data science for exp. life sciences with R (pt. 2) — Forsch.Einr. BMEL via zoom
- 2024 Jun — Data Science in den exp. Naturwiss. (Tl. 2) — Forsch.Einr. BMEL via zoom
- 2024 Apr — Data science for exp. life sciences with R (pt. 1) — Forsch.Einr. BMEL via zoom
- 2024 Apr — Data Science in den exp. Naturwiss. mit R (Tl. 1) — Forsch.Einr. BMEL via zoom
- 2024 Apr — Data Science with R (pt. 2) — Max Planck Inst. Tübingen via zoom
- 2024 Apr — Data Analytics mit Python — Karrierecenter der Bundeswehr (e-learning)
- 2024 Mar — Data Analytics mit Python — Karrierecenter der Bundeswehr (e-learning)
- 2024 Feb — Advanced data visualization in R — 70th Biometrical Colloquium, Lübeck

2023

- 2023 Dec — Feldversuche und Statistik - Interaktive Beratung — Hochschule Nürtingen-Geislingen via zoom
- 2023 Dec — Data science for exp. life sciences with R (pt. 2) — Forsch.Einr. BMEL via zoom
- 2023 Dec — Data Science in den exp. Naturwiss. (Tl. 2) — Forsch.Einr. BMEL via zoom
- 2023 Dec — Statistics with R - an Introduction — Universität Bonn via zoom
- 2023 Nov — exp. Design - Practicals in R — CIHEAM Zaragoza via zoom
- 2023 Nov — Data Science with R - an Introduction — Max Planck Inst. Tübingen via zoom
- 2023 Oct — Data science for exp. life sciences with R (pt. 1) — Forsch.Einr. BMEL via zoom
- 2023 Oct — Data Science in den exp. Naturwiss. mit R (Tl. 1) — Forsch.Einr. BMEL via zoom
- 2023 Jul — R Introduction — Universität Flensburg via zoom
- 2023 Jul — Statistics with R - an Introduction — Universität Bonn via zoom
- 2023 Jun — Data science for exp. life sciences with R (pt. 2) — Forsch.Einr. BMEL via zoom
- 2023 Jun — Data Science in den exp. Naturwiss. mit R (Tl. 2) — Forsch.Einr. BMEL via zoom
- 2023 May — Statistics with R - an Introduction — Universität Bonn via zoom
- 2023 May — Data science for exp. life sciences with R (pt. 1) — Forsch.Einr. BMEL via zoom
- 2023 May — Data Science in den exp. Naturwiss. mit R (Tl. 1) — Forsch.Einr. BMEL via zoom
- 2023 Feb — Introduction to data science for exp. life sciences with R — Pro-RUWA via zoom

2022

- 2022 Nov — Data science for exp. life sciences with R (pt. 2) — Forsch.Einr. BMEL via zoom
- 2022 Nov — Data Science in den exp. Naturwiss. mit R (Tl. 2) — Forsch.Einr. BMEL via zoom
- 2022 Nov — Data science for exp. life sciences with R (pt. 1) — Forsch.Einr. BMEL via zoom
- 2022 Nov — Data Science in den exp. Naturwiss. mit R (Tl. 1) — Forsch.Einr. BMEL via zoom
- 2022 Nov — Statistics with R - an Introduction — Universität Bonn via zoom
- 2022 Oct — R and the {Tidyverse} — FBN, Dummerstorf via zoom
- 2022 Mar — Data science for exp. life sciences with R (pt. 2) — Forsch.Einr. BMEL via zoom
- 2022 Mar — Data Science in den exp. Naturwiss. mit R (Tl. 2) — Forsch.Einr. BMEL via zoom
- 2022 Mar — Data science for exp. life sciences with R (pt. 1) — Forsch.Einr. BMEL via zoom
- 2022 Mar — Data Science in den exp. Naturwiss. mit R (Tl. 1) — Forsch.Einr. BMEL via zoom

2021

- 2021 Dec — Statistics with R (Beginner) — Universität Kassel
- 2021 Jul — Data science in den Naturwiss. mit R (Tl. 2) — Forsch.Einr. BMEL via zoom
- 2021 May — Data science in den Naturwiss. mit R (Tl. 1) — Forsch.Einr. BMEL via zoom
- 2021 Mar — Data science in den Naturwiss. mit R (Tl. 2) — Forsch.Einr. BMEL via zoom

2020

- 2020 Nov — Planning exp. designs, repeated meas., and their analyses in R — Universität Kassel via zoom

- 2020 Nov — Data science in den Naturwiss. mit R (Tl. 1) — Forsch.Eintr. BMEL via zoom
- 2020 Oct — exp. Design - Practicals in R — CIHEAM Zaragoza via zoom
- 2020 Mar — Real-time consultation on statistics and mixed models in R — Universität Kassel

2019

- 2019 Dec — Basics of applied statistics — Universität Rostock
- 2019 Nov — Data science in den Naturwiss. mit R (Tl. 2) — Forsch.Eintr. BMEL, Braunschweig
- 2019 Oct — Data science in den Naturwiss. mit R (Tl. 1) — Forsch.Eintr. BMEL, Braunschweig
- 2019 Sep — Essential basics of statistics — Universität Rostock

2018

- 2018 Nov — Gemischte Modelle in R — Forsch.Eintr. BMEL, Braunschweig
- 2018 May — Implementation of yield stability assessment with ASReml-R — Bangladesh Rice Res. Inst., Gazipur

2016

- 2016-2018 — Statistical analysis with SAS (monthly) — Universität Hohenheim, Stuttgart
- 2016-2018 — Statistical analysis with R (monthly) — Universität Hohenheim, Stuttgart

Scientific Publications

Buntaran, H., Piepho, H.-P., Schmidt, P., Rydén, J., Halling, M., & Forkman, J. (2020). Cross-validation of stagewise mixed-model analysis of swedish variety trials with winter wheat and spring barley. *Crop Science*, 60(5), 2221–2240. <https://doi.org/10.1002/csc2.20177>

Friedrichs, P., Schmidt, P., & Schmidt, K. (2021). *Protanopie und protanomalie bei berufskraftfahrern und berufskraftfahrerinnen – prävalenz und unfallrisiko = protanopia and protanomaly among professional drivers: Prevalence and accident risk: Vols. Heft 319*. BioMath GmbH; Bundesanstalt für Straßenwesen; Fachverlag NW in der Carl Schünemann Verlag GmbH. <https://bast.opus.hbz-nrw.de/frontdoor/index/index/searchtype/series/id/5/start/1/rows/25/docId/2574>

Kukowski, S., Schmidt, P., Piepho, H.-P., Röhl, M., Hauffe, H.-K., & Streck, T. (2020). Auswirkungen atmosphärischer stickstoffeinträge auf magere flachland-mähwiesen in baden-württemberg. *Natur Und Landschaft*, 95(2), 58–67. <https://doi.org/10.17433/2.2020.50153773.58-67>

Rahman, N. Md. F., Malik, W. A., Kabir, Md. S., Baten, Md. A., Hossain, Md. I., Paul, D. N. R., Ahmed, R., Biswas, P. S., Rahman, Md. C., Rahman, Md. S., Iftekharuddaula, K. Md., Hadasch, S., Schmidt, P., Islam, Md. R., Rahman, Md. A., Atlin, G. N., & Piepho, H.-P. (2023). 50 years of rice breeding in bangladesh: Genetic yield trends. *Theoretical and Applied Genetics*, 136(1), 1432–2242. <https://doi.org/10.1007/s00122-023-04260-x>

Schmidt, K., Friedrichs, P., Cornelsen, H. C., Schmidt, P., & Tischer, T. (2021). *Musculoskeletal disorders among children and young people: Prevalence, risk factors, preventive measures – a scoping review*. European Agency for Safety; Health at Work. <https://doi.org/10.2802/511243>

Schmidt, K., Friedrichs, P., & Schmidt, P. (2022). *Warenstromanalyse tierischer lebensmittel: Gutachten zur erzeugung, verarbeitung, vermarktung und zum verzehr von fleisch, milch und eiern in deutschland (158/2022)*. Umweltbundesamt. https://www.umweltbundesamt.de/sites/default/files/medien/479/publikationen/texte_158-2022_warenstromanalyse_tierischer_lebensmittel.pdf

Schmidt, K., Schmidtke, J., & Schmidt, P. (2020). *Studie zum potenzial von wirtschaftsdünger zur energetischen verwertung im land brandenburg. schlussbericht (Az.: VV-0039-2017)*. Ministerium für Ländliche Entwicklung, Umwelt und Landwirtschaft (MLUL) des Landes Brandenburg. <https://mluk.brandenburg.de/sixcms/media.php/9/Potenzial-Wirtschaftsduenger-energetische-Verwertung-Brandenburg.pdf>

Schmidt, K., Schmidtke, J., Schmidt, P., Kohl, C., Wilhelm, R., Schiemann, J., Voet, H. van der, & Steinberg, P. (2017). Variability of control data and relevance of observed group differences in five oral

toxicity studies with genetically modified maize MON810 in rats. *Archives of Toxicology*, 91(4), 1977–2006. <https://doi.org/10.1007/s00204-016-1857-x>

Schmidt, P. (2019). *Estimating heritability in plant breeding programs* [Dissertation to obtain the doctoral degree of Agricultural Sciences (Dr. sc. agr.), University of Hohenheim; Biostatistics Unit (340c), Institute of Crop Science (340)]. <http://opus.uni-hohenheim.de/volltexte/2020/1720/>

Schmidt, P., Hartung, J., Bennewitz, J., & Piepho, H.-P. (2019). Heritability in plant breeding on a genotype-difference basis. *Genetics*, 212(4), 991–1008. <https://doi.org/10.1534/genetics.119.302134>

Schmidt, P., Hartung, J., Rath, J., & Piepho, H.-P. (2019). Estimating broad-sense heritability with unbalanced data from agricultural cultivar trials. *Crop Science*, 59(2), 525–536. <https://doi.org/10.2135/cropsci2018.06.0376>

Schmidt, P., Möhring, J., Koch, R. J., & Piepho, H.-P. (2018). More, larger, simpler: How comparable are on-farm and on-station trials for cultivar evaluation? *Crop Science*, 58(4), 1508–1518. <https://doi.org/10.2135/cropsci2017.09.0555>

Tulinská, J., Adel-Patient, K., Bernard, H., Líšková, A., Kuricová, M., Ilavská, S., Horváthová, M., Kebis, A., Rollerová, E., Babincová, J., Aláčová, R., Wal, J.-M., Schmidt, K., Schmidtke, J., Schmidt, P., Kohl, C., Wilhelm, R., Schiemann, J., & Steinberg, P. (2018). Humoral and cellular immune response in wistar han RCC rats fed two genetically modified maize MON810 varieties for 90 days (EU 7th framework programme project GRACE). *Archives of Toxicology*, 92(7), 2385–2399. <https://doi.org/10.1007/s00204-018-2230-z>

Zeljenková, D., Aláčová, R., Ondřejková, J., Ambušová, K., Bartušová, M., Kebis, A., Kovřížnych, J., Rollerová, E., Szabová, E., Wimmerová, S., Černák, M., Krivošíková, Z., Kuricová, M., Líšková, A., Spustová, V., Tulinská, J., Levkut, M., Révájová, V., Ševčíková, Z., ... Steinberg, P. (2016). One-year oral toxicity study on a genetically modified maize MON810 variety in wistar han RCC rats (EU 7th framework programme project GRACE). *Archives of Toxicology*, 90(10), 2531–2562. <https://doi.org/10.1007/s00204-016-1798-4>